Energy and Buildings 43 (2011) 179–188

Contents lists available at ScienceDirect

Energy and Buildings

journal homepage: www.elsevier.com/locate/enbuild

An embodied carbon and energy analysis of modern methods of construction in housing: A case study using a lifecycle assessment framework

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ARTICLE INFO

Article history:
Received 27 November 2009
Received in revised form 25 August 2010
Accepted 11 September 2010

Keywords:
Life cycle assessment (LCA)
Housing
Residential buildings
Construction
Embodied carbon
Embodied energy
Modern methods of construction

ABSTRACT

There is a growing interest in comparing the energy and consequential carbon embodied in buildings using different methods of construction and alternative materials. This paper compares the embodied carbon in a low energy, affordable house constructed using a novel offsite panellised modular timber frame system, in Norfolk UK with two traditional alternative scenarios. A lifecycle assessment (LCA) framework is used to conduct a partial LCA, from cradle to site, of the construction. An inventory of the materials and fossil fuel energy utilised in the construction was used to calculate the primary energy consumed and the associated embodied carbon. The embodied carbon was found to be 34.6 tonnes CO_2 for a 3 bedroom semi-detached house, 405 kgCO_2 per m^2 of useable floor area. When compared with traditional methods of construction the modern methods of construction (MMC) house resulted in a 34% reduction in embodied carbon. Despite timber being the predominant structural and cladding material, concrete is the most significant material (by proportion) in embodied carbon terms, responsible for 36% of materials related embodied carbon.

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1. Introduction and back ground

"The scientific evidence is now overwhelming: climate change presents very serious global risks, and it demands an urgent global response" [1].

Climate change, the most serious threat to human society, is, ironically, a threat that human society has created itself. Global atmospheric concentrations of greenhouse gases (GHGs) have increased since 1750, notably carbon dioxide (CO_2) the most predominant greenhouse gas by volume. Emissions of CO_2 from fossil fuel combustion, in conjunction with that emitted from cement manufacture, are responsible for more than 75% of the increase in atmospheric CO_2 since the pre-industrial 18th century [2].

The construction and occupation of buildings is a substantial contributor of global CO_2 emissions, with almost a quarter of total global CO_2 emissions attributable to energy use in buildings [3]. A further 5% can be attributed to the manufacture of cement, a principal construction material [4]. Reducing the energy demand and consequential carbon emissions attributed to buildings is clearly an important goal for government climate policy.

The energy and associated emissions of carbon (referred to in terms of CO_2) linked with the lifecycle of buildings can be considered in three distinct, but inter-linked stages. These are construction, occupation and end of life deconstruction. Reducing occupational energy has been a significant focus for UK mitigation policy at national level, in particular that relating to reducing the energy demand of housing. Although the energy used and consequential carbon emitted during the occupation of a building equates to the majority of that buildings lifetime carbon footprint, there are significant carbon consequences involved in the initial construction of a building.

The extraction, processing, manufacture, transportation and use of a product utilises energy and produces many environmental impacts, including emissions of CO_2. With the exception of the generally more evident energy in use, these impacts are regarded as the hidden, or embodied, burdens. Embodied energy and carbon are not, in general practice, a consideration when a building is designed, specified and constructed.

For housing constructed to conventional standards embodied energy is equivalent to a few years of operating energy, although there are exceptions to this, such as low energy buildings [5]. Embodied carbon is of particular importance for low energy buildings [6] because although less energy is used during occupation, additional energy is often required for the manufacture of the increased levels of insulation, the heavier mass materials used and the additional technologies often deployed. The embodied carbon of a low energy house is likely to contribute a greater proportion of its overall lifecycle carbon emissions during that buildings lifetime than would occur for a conventional house. The size of this substitution effect is unclear. It has been suggested that between 2 and

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doi:10.1016/j.enbuild.2010.09.005

36% of a traditional house lifetime energy demand is attributable to the winning of primary materials, manufacture, transport and construction of a building [7]. This range increases to 9–46% for a low energy house. The materials specified and the construction technologies used greatly influence the overall embodied energy and carbon emissions during the construction phase.

Another area of concern is the embodied energy associated with waste. In the UK the construction industry is responsible for over a third of all waste arisings [8], 51% of which is recycled or reused, the majority as aggregate [9]. On site construction typically has contingency and error related over ordering, amounting to approximately 10% of all materials brought to site, with 10–15% of the materials imported to a construction site being exported as waste [10]. Reducing the embodied energy of construction needs to address both the efficiency of manufacture and the efficiency of use. One solution to this problem is the increased use of offsite manufacturing, or modern methods of construction (MMC), of housing components or whole houses. The factory production of construction elements can have much lower resource inputs and reduced waste outputs than compared with on-site construction [11]. A recent report estimated the waste reduction through substitution of traditional methods with prefabrication systems to be between 20 and 40%, the greater the prefabrication the greater the savings [11].

The UK is committed to increasing the number of new houses, 3 million by 2020, and a substantial increase in the thermal renovation of the existing housing stock [12]. This increase in construction will have significant implications for the UK's national carbon budget. However the magnitude of this impact will be dependent upon how these houses are constructed.

This paper contributes to the growing literature on the environmental impacts of construction by comparing the embodied energy and carbon consequence of the construction of a novel low energy offsite modular timber framed house with an identical house constructed using traditional methods and assesses whether MMC construction can contribute towards national carbon reduction goals. The objectives of the study are to:

- Identify and quantify the embodied energy in the construction of a low energy house constructed using MMC
- Quantify the embodied carbon in the construction of an MMC house
- Compare the study model with traditional methods of construction
- Quantify the potential national carbon savings from the expansion of the use of MMC construction

The results of this research contribute to understanding the carbon emissions from housing that are all too often overlooked.

This paper is organised as follows. Section 2 gives a brief overview of the literature specific to LCA studies of housing in the UK and summarises the findings of these studies in terms of GJ per m^2 and, or, $kgCO_2$ per m^2 of floor area. The case study and methodology are described in Sections 3 and 4. The results of the inventory analysis of the case study house (scenario 1) are given (Section 5) before a summary of the two alternative scenarios (scenarios 2 and 3) in Section 6. The results are discussed in Section 7 before an discursive exploration of the implications for the UKs house building programme and its impacts on national carbon emissions (Section 7). The paper concludes in Section 8.

2. A review of housing LCA literature

There is a growing body of literature on embodied energy and carbon in the construction of houses. Studies typically use a process based LCA methodology (bottom up) rather than an input–output (top-down) methodology. Individual process based studies have used different parameters, factors, datasets and boundaries. In addition, values of embodied energy, and consequential emissions of carbon, vary by country due to: the energy mix; transformation processes; the efficiency of the industrial and economic system of that country; and how these factors vary over time [7]. Consequently, results from lifecycle studies are indicative and should be interpreted with caution and careful attention to the methods used, the system boundaries applied, what has (or has not) been included before any interpretation can be made or conclusions drawn.

Many early studies are principally concerned with embodied energy, expressed as primary energy. Nässén et al. [13] summarised the results of 20 process based (predominantly Scandinavian) studies published prior to 2001. The studies showed similar results with a range of 1.3–7.3 GJ/m^2 primary energy for residential buildings.

Literature specific to embodied carbon and energy of UK housing construction is sparse. Table 1 shows a summary of the literature specific to the UK.

In a recent study specific to UK housing construction, Hammond and Jones [14] reported an average of 5.3 GJ per m^2 embodied energy and 403 $kgCO_2/m^2$ embodied carbon from 14 predominantly UK process based case studies using an open access inventory of carbon and energy data for a comprehensive range of construction materials. The average embodied energy is comparable with the findings of Nässén et al. [13].

Other studies relevant to the UK give incomplete detail of the system under study and use older embodied energy data [15,16]. Asif et al. [15] reported 3.25 GJ/m^2, which is at the lower end of the range found in previous studies. Hacker et al. [16] in a comparison of lightweight (timber frame) to heavyweight (concrete) found a range of embodied carbon 492–569 $kgCO_2/m^2$ but did not give findings as primary energy.

The majority of the studies cited are not comparative, lack the level of detail required to make any comparisons and have inconsistent boundaries. Despite this, there is a consistent range of embodied energy and carbon results to be found within the literature.

None of the studies cited have quantified and compared the carbon from construction of housing constructed using MMC methods with that from conventional construction in the UK. This forms the basis of the study detailed in this paper, in which the embodied carbon of a house constructed using MMC methods is compared with two alternative construction scenarios for the same house using a life cycle framework. The results are considered in the context of the UKs house building programme and its impacts on national emissions targets.

3. Description of the case study

3.1. Model house

The case study presented here (scenario 1) is based on a low energy affordable house constructed in 2008 in Lingwood, Norfolk in the UK. A three bedroom semi-detached house of 83 m^2 internal floor area was used as the model for the case study. In addition to the case study two further scenarios (scenarios 2 and 3) were modelled to provide a comparison. The dimensions and design specifications are given in Table 2 and Fig. 1.

3.1.1. Scenario 1: MMC timber frame larch cladding

The scenario 1 house was constructed using a novel approach which combined offsite modular timber frame system with additional insulation materials to exceed current mimimum building regulation standards and untreated Siberian larch weather board-

Table 1
An overview of literature specific to UK housing construction embodied energy and carbon case studies.

Source	Case numbers	LCA scope	Construction	GJ/m^2	kgCO$_2$/m^2
Hammond and Jones [14]	14	Cradle to construction	Mixed	5.34	403
Asif et al. [15]	1	Cradle to construction	Unknown	3.25	x
Hacker et al. [16]	4	Cradle to occupation	Mixed	x	492–569

ing installed onsite. The use of timber as a facade material is becoming more prevalent in commercial buildings as an aesthetic nod towards a buildings sustainable credentials but is still uncommon in mass produced housing in the UK.

The pre-manufactured timber frame is a factory constructed modular system consisting of wall modules of a softwood timber frame with factory installed phenolic foam insulation to meet minimum building regulation standards, cement wall board and a waterproof membrane (Fig. 2). The modules are constructed to enable assembly on site with the addition of design flexibility in the material used as a facade. The first floor modules are constructed using engineered timber (I-beams and Glulam beams).

The substructure, foundations, first floor and roof are constructed using traditional approaches. The substructure consists of an over site poured concrete slab with steel reinforcement in shallow strip footings. Brick and block walling are used below the damp proof course. The suspended ground floor is formed from precast steel reinforced beams with an infill of concrete blocks.

3.1.2. Scenario two: MMC timber frame brick cladding

In the scenario 2 house brick cladding replaces the larch cladding and its associated components. The substitution required an increase in the wall width of the model house (Table 2). No other parameters were altered.

3.1.3. Scenario three: conventional masonry cavity wall

In the scenario 3 house the timber frame and larch cladding is replaced by a traditional masonry construction. This consists of a block internal wall, a cavity filled with a phenolic insulation and an outer brick cladding. Steel ties were assumed to tie the inner and outer walls together. In this scenario the model was affected by: an increased wall width; increased substructure to accommodate the additional mass and wall width.

Table 2
Design parameters of the three case study scenarios: (1): MMC timber frame with larch cladding; (2) MMC timber frame with brick cladding; (3) conventional masonry cavity wall.

	(1) MMC timber frame larch cladding	(2) MMC timber frame brick veneer	(3) Conventional masonry cavity wall
Number of floors	2	2	2
Total internal floor area (m^2)	91	91	91
Total footprint area (m^2)	45.3	46.6	46.8
Total wall area (m^2)	113.5	115.0	115.3
Wall width (m^2)	273	319	327
Opening area (m^2)	16	16	16
Framework	Timber	Timber	Masonry
u-Value (Wm^2 k):			
Wall	0.18	0.18	0.18
Floor	0.16	0.16	0.16
Roof	0.14	0.14	0.14
Windows	1.80	1.80	1.80

4. Methododology

4.1. Life cycle assessment framework

The growing importance of environmental issues, such as climate change, has created a need to evaluate the impacts of the products we use. One of the principle techniques to enable the quantification and comparison of the environmental impacts of a product is life cycle assessment (LCA). LCA is a framework for evaluating the environmental impacts of a product, process or service from cradle to grave and is carried out according to International Standards, ISO 14040 [17]. ISO 14040 defines LCA as the:

'Compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product system throughout its life cycle.'

The LCA framework consists of four main phases [17]:

1. Goal, scope and definition
2. Inventory analysis (LCI)

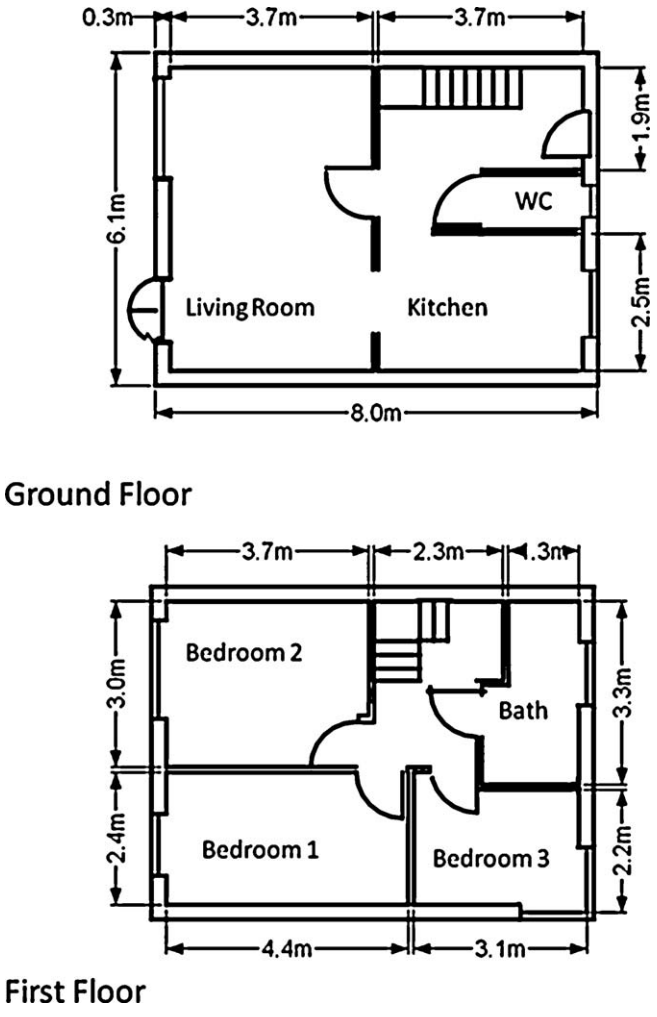


Fig. 1. Plan dimensions of case study house.

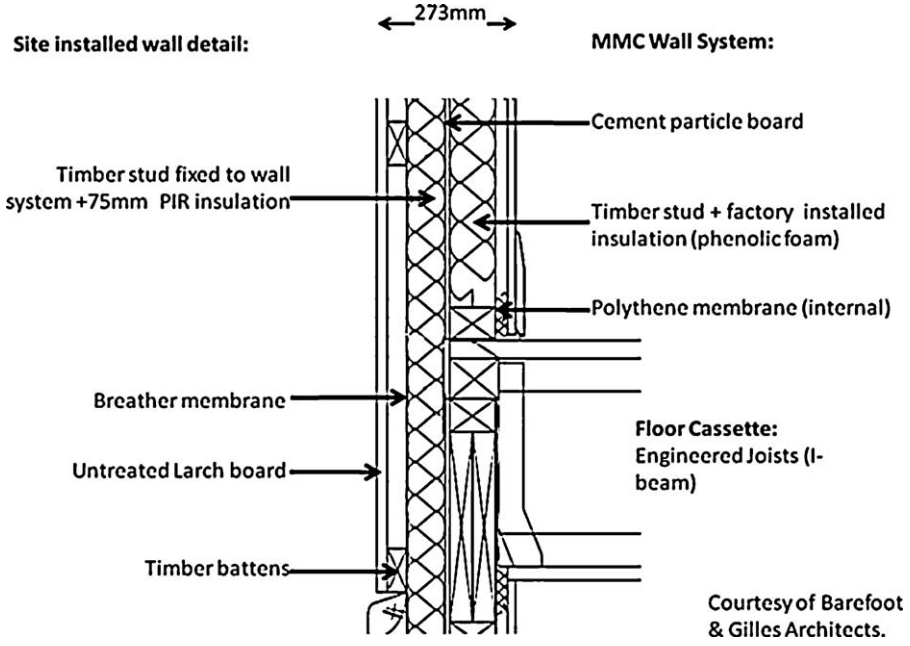


Fig. 2. Sketch of MMC wall and floor components.

Barefoot and Gilles Architects 2008

3. Impact assessment (LCIA)
4. Interpretation.

The initial phase defines the scope of the study, including defining the functional unit, the system boundary, level of detail and how the environmental burdens will be allocated. It is dependent upon the subject and the intended use of the study and can vary considerable depending upon the particular LCA [17].

The second phase, the life cycle inventory (LCI), is the compilation of an inventory of the input/output data with regard to the system under study. It is an iterative process, with data constantly being updated, added to as more is learnt about the system under study.

The third phase, the impact assessment (LCIA), evaluates the significance of potential environmental impacts using the LCI results and provides information for the final interpretation phase.

Life cycle interpretation is the final phase in the LCA framework. The results of an LCIA (or an LCI in a partial LCA study) are summarised to form a basis for conclusions, recommendations and decision making in accordance with the study as defined in the first phase [17].

4.2. Goal of study

The study presented in this paper uses an LCA framework as a tool to conduct a partial LCA, from cradle to site of the construction of a low energy house constructed using an offsite panellised modular timber frame system (Fig. 3). An inventory of the materials involved in the construction and the fossil fuel energy used during the construction is used to calculate the primary energy used and the associated embodied carbon. The goal of the study is to investigate the carbon consequences of constructing new housing, comparing different approaches and identifying areas that could deliver reductions in embodied carbon.

4.3. Case study boundaries:

The study scope includes the cradle to site emissions from:

- materials and products used in construction
- final transport of the materials and products to site
- materials waste produced on site
- transportation of waste to disposal
- fossil fuel energy used on site during construction and in manufacture of MMC components

The infrastructure required in production, such as roads, factories, warehouses and machinery, and the operational activities associated with administration and the workforce themselves (including their transport to site) were outside the boundaries of the study and are excluded.

The environmental impacts, where a process produces multiple or subsidiary products (such as timber production at a sawmill producing sawdust, woodchip and bark for use in wood fibre board manufacture or as fuels), are allocated by mass.

In order to be able to make a fair comparison between different materials and approaches used a unit of study has to be defined (termed a functional unit). The functional unit for this study is the external, thermal envelope of a 3 bedroom, semi-detached house with a total foot print area of 45 m^2 and a total internal volume of 220.5 m^3. For the purposes of the study internal finishes and fittings are excluded. The study assumed these would be equivalent for all construction types and, therefore, outside the scope of this study.

4.4. Inventory and data sources

The inventory of materials and inputs into the construction of the development were estimated from information provided by the quantity surveyors, architects, contractors and companies providing goods and services along the supply chain for this property. The data were collected retrospectively, with varying degrees of quality as discussed below.

The dimensions of the house were obtained from the architects plans.

Material quantities were obtained from quantity survey data, plans and information provided by supply chain partners. In cases, such as shared party walls, the materials were allocated by area.

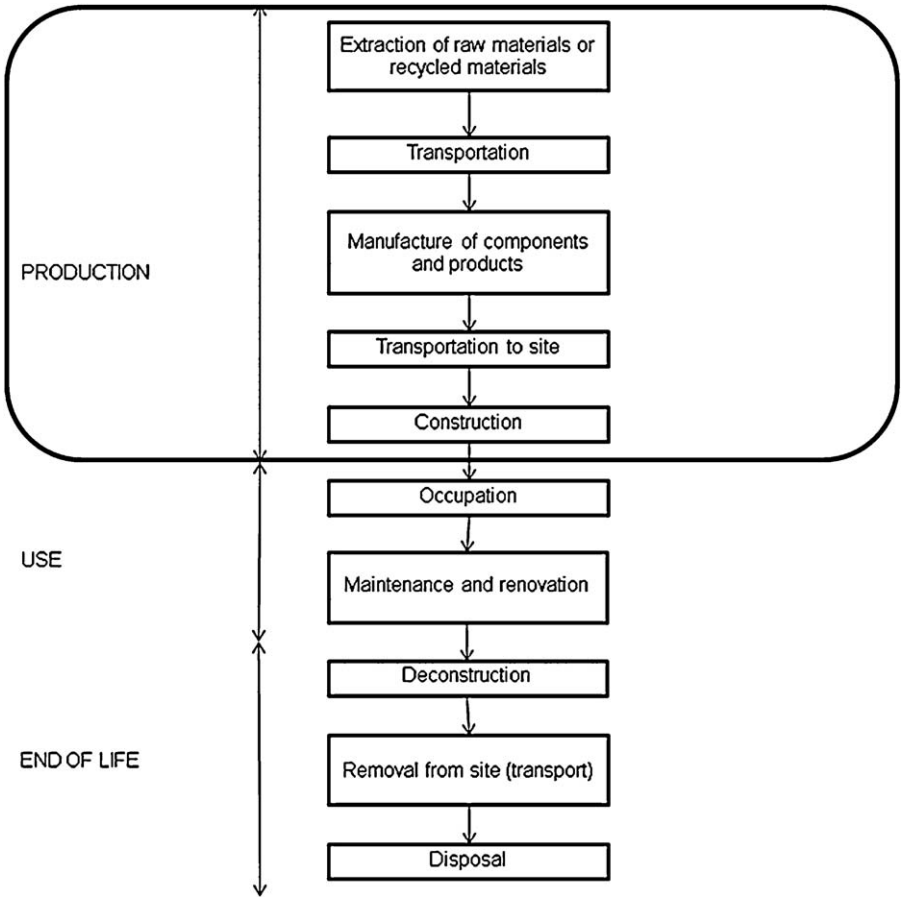


Fig. 3. A simplified lifecycle process flow chart showing production boundary for the case study.

Information regarding the offsite frame production process was obtained from the manufacturing company who provided aggregated production data on energy and materials. Some data were unavailable due to commercial confidentiality, in particular pertaining to the insulation. Data gaps were filled with published literature references where available or best guess estimates. Allocation of energy and waste from the manufacturing process was by units of production.

All the timber materials were imported. The larch cladding was imported by boat from the Irkutsk region of Siberia. The timber softwood was imported from Scandinavia. The structural engineered timber was produced and imported from the United States.

No detailed records on waste were kept during the construction process, with the data on waste generated during on-site construction being limited to an aggregated volume. Therefore estimates of different waste streams and disposal routes were made based on benchmark data from The Smart Waste Scheme [18] and from published literature [19,20,11]. Limited information on waste management practices and material separation was obtained from observation on site, the site operators and waste management contractors.

Energy and fuels used onsite during construction, including petrol, diesel, gas and electricity, were derived from receipts and meter readings. It was not possible to disaggregate the energy consumed to specific activities and, therefore, specific build components. Onsite energy is therefore presented as an aggregated figure for electricity and each fuel; more detailed analysis is beyond the scope of this paper.

Carbon emissions factors and embodied energy factors for materials, processes and fuels were derived where possible from the UK or relating to the country of production. A number of sources and databases were used including:

- published Government carbon emission factors [21]
- The Inventory of Carbon and Energy [14]
- Ecoinvent [22]
- U.S. Life-Cycle Inventory (USLCI) [23]

Simapro V7.1 software was used in the analysis of the engineered timber components using the above inventory databases. Simapro (PRé Consultants, Amersfoort, The Netherlands) is a dedicated LCA software tool for undertaking LCA studies.

5. Results: inventory analysis

5.1. Scenario 1: MMC timber frame larch cladding

The case study house requires a total of 519 GJ of primary energy to construct, which equates to an embodied primary energy of approximately 5.7 GJ per m^2 of floor area (Table 3). The carbon embodied in the construction of the house amounts to 34.6 tonnes CO_2, approximately 405 $kgCO_2$ per m^2 of useable floor area (Table 3). Further details of the results of the lifecycle inventory and the embodied energy and carbon values of the materials used are summarised in Table 3 and Fig. 4. The remainder of this paper presents the results in terms of carbon as kg or tonnes of CO_2.

82% of the total embodied carbon is embodied in the materials incorporated in the building (exclusive of waste). The remainder were attributed to construction activities such as transporting

Table 3
Summarised inventory of materials, transport and fuels used in the construction of the scenario 1 house and associated primary energy and embodied CO_2.

Category	Material	Quantity (kg)	Emissions ($kgCO_2$)	Primary energy (MJ)
Metals	Aluminium	260	2140	40,260
	Steel	251	956	10,722
Minerals	Brick	2264	1175	18,510
	Cement (mortar/board)	2023	798	12,997
	concrete	56,651	9863	72,142
	Gypsum plaster products	1349	413	7207
Openings	Windows	1277	1996	40,584
	Doors	142	246	4624
Plastics	HD polyethylene	56	90	4330
	LDPE	29	72	2558
	Polyisocyanate insulation	187	561	13,477
	Polythene	146	285	13,152
	PUR insulation	195	585	14,058
Timber	Composite board products	4330	3462	64,116
	Larch	1315	1421	15,090
	Engineered timber	222	152	2811
	Softwood	6792	3056	50,262
Fuel	Mains gas (kWh)	1107	226	4128
	UK Grid electricity (kWh)	11,106	948	12,462
	Diesel (l)	2070	363	5328
Waste		5350	4934	96,728
Transport	Factory gate to site	9372 tkm	883	13,131
Total:			34,625	518,677

materials from point of distribution to site, waste materials exported from the site and energy used onsite. Concrete and waste are the two predominant groups (Fig. 4).

In considering materials (Fig. 5) minerals are the most significant material category, accounting for 45% of material related embodied carbon (excluding waste materials). The minerals category includes materials such as cement, gravels, sands and concrete products. Concrete is the main contributor, with 36% of the embodied carbon associated with materials being derived from concrete (Figs. 4 and 5). Much of this is due to Portland cement, which has a high embodied energy of 0.83 $kgCO_2$ per kg of product at the factory gate of the cement works in the UK [14].

The majority of minerals were used in the construction of the substructure and foundations. These elements were responsible for 71% of the emissions associated with the use of minerals. The remainder of the minerals were incorporated in the ground floor (concrete block and beam, 16% of minerals emissions) and the roofing tiles (concrete tiles, 9% of minerals emissions).

As illustrated in Fig. 6 half of the embodied carbon was attributable to the substructure, foundations and ground floor. The principle material in these construction elements is minerals,

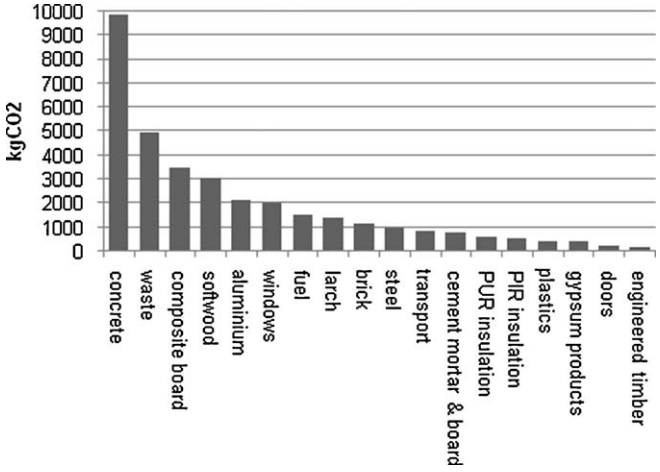


Fig. 4. Summarised inventory results showing embodied carbon in construction, inclusive of offsite frame manufacture ($kgCO_2$).

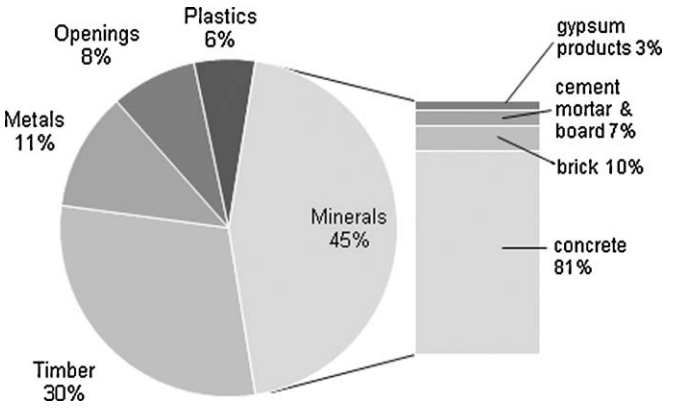


Fig. 5. Proportion of embodied carbon in materials (excluding waste, transport and energy in construction).

specifically concrete. The MMC frame was responsible for 12% of the embodied carbon.

Timber, a key material in the structure and external cladding, was responsible for 30% of the total emissions (inclusive of the timber used in the offsite frame and transport to UK distribution points). The majority of timber was found in the walls and

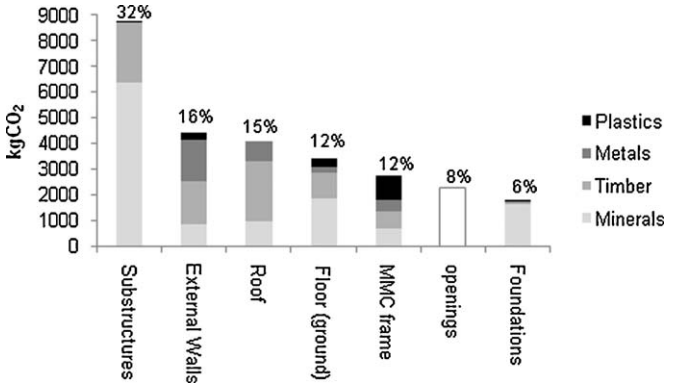


Fig. 6. Proportions of material embodied carbon attributed to each structural component.

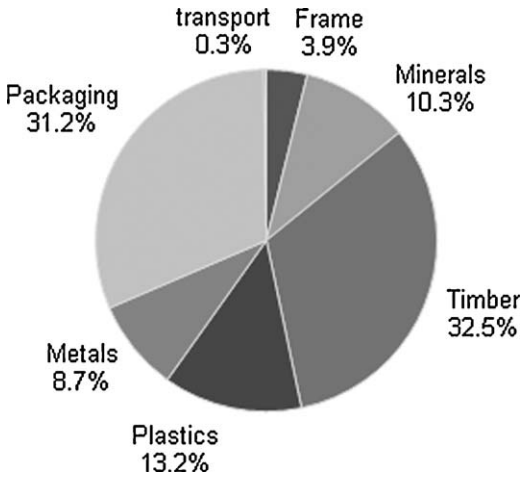


Fig. 7. Proportions of different types of waste (by weight) occurring both onsite and offsite during the manufacture of the timber frame (off-site waste shown as frame category).

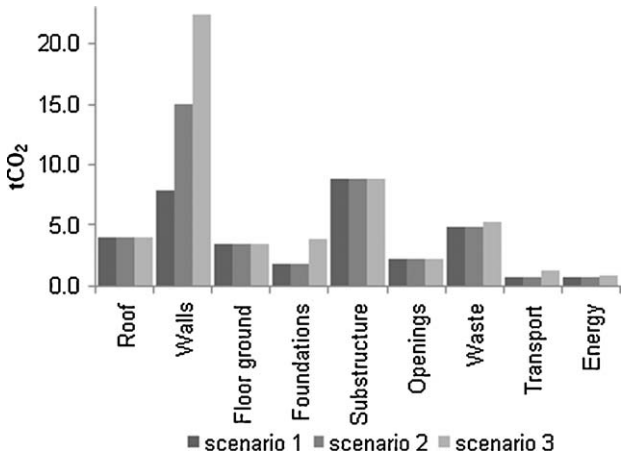


Fig. 8. Comparison of inventory results by component for scenario 1, scenario 2 and scenario 3.

roof. The larch cladding was responsible for 5% of all material emissions.

The remaining 25% of material related embodied carbon was attributable to metals, the openings (doors and windows) and plastics.

5.2. Transportation

Transportation from factories and distribution points to the site accounts for 9372 tkm, resulting in 2% of the total embodied carbon. This is similar to other studies which also found transport to have a relatively low share of the total emissions of CO_2 [24].

5.3. Waste

The construction waste consisted of two main waste streams, that occurring during onsite construction and that occurring during manufacturing of the frame.

In total 17 m^3 of waste materials were reported to have been exported from the site. This included excavated inert materials, waste and unused construction materials and other waste. An estimated 4.9 tCO_2 resulted from this waste, equating to 109 $kgCO_2$ per m^2. Timber and packaging were the predominant contributors (33% and 31%, respectively) (Fig. 7). It was estimated that 65 kg of waste (excluding inert site excavation materials exported from site) were produced for each m^2 of floor area.

The manufacturing of the frame contributed the least to waste related embodied carbon. However the manufacturing waste data was incomplete, in particular relating to packaging, the majority of which was reported as being recycled. The quantity of materials accounted for are given in Table 4. It was estimated that the reduction in embodied carbon attributed to resource efficiency was 0.3 tCO_2 [11]. During the offsite manufacturing process, production waste was either returned to the manufacturing process or, being produced in quantities that are viable for export offsite, recycled into other alternative processes and products, resulting in just 2 kg of waste produced for each m^2 of internal floor area.

Table 4
Summarised waste materials from offsite manufacture of MMC structural components.

Waste category	Material	Quantity (kg)	Embodied carbon ($kgCO_2$)
Reused	Timber	53	24
Landfill	Total landfill waste	140	168
	Insulation	33	100
	Cement particle board	47	24
	Sawdust		
	Timber sawdust	40	18
	Plastic	13	25
	Transport (t/km)	7	0.9

6. Results—alternative scenarios

6.1. Scenario 2 MMC timber frame brick cladding

The brick clad house scenario had an embodied carbon of 45.6 tCO_2 and required 656 GJ of primary energy. This equates to 535 $kgCO_2$ m^2 and 7.7 GJ per m^2 primary energy per usable floor area, increases of 32% embodied carbon and 35% embodied energy compared to the case study model house of scenario 1.

The walls (including brick veneer and frame) in this scenario are responsible for 41% of the total embodied carbon, compared with 23% for scenario 1 (Fig. 8). Of this the non-frame elements (brick, additional insulation, membranes, etc.) formed 81% with the offsite frame responsible for the remainder. The substructure and foundations are accountable for 23% of the total emissions.

Unsurprisingly the majority of this difference is accounted for by the increase in minerals (i.e. brick, cement and sand) associated with the brick cladding. Typically, an increase in the total proportion of heavier materials in a construction project will increase transport emissions, in this case by 25%. There was also a 14% increase in construction energy due to the increase in machinery required on site, such as mixers for the mortar.

6.2. Scenario 3: masonry

Scenario 3 considers the implications of using a traditional masonry construction consisting of a brick, insulated cavity and block wall.

Scenario 3 was found to have embodied carbon of approximately 52 tCO_2, and required 700 GJ energy. This equates to 612 $kgCO_2$ m^2 and 8.2 GJ per m^2 primary energy per usable floor area, increases of 51% embodied carbon and 35% embodied energy compared to the case study model house of scenario 1.

The majority of this difference is attributed to the differences in embodied carbon in the walls and foundations (Fig. 8). In this scenario 67% of the total embodied carbon was accounted for by the walls, foundations and substructure, 43% and 24%, respectively. Masonry construction requires an increased volume in load bearing foundations to accommodate the heavier masonry walls.

Materials accounted for 86% of the total embodied carbon. 77% of which was attributed to minerals.

7. Discussion

This study found that a house constructed using a panellised timber frame MMC construction, produced a building with a 34% reduction in embodied carbon when compared with a traditional masonry construction for an equivalent house.

The reduction in the use of materials with relatively high embodied carbon, in this case brick and blocks, with materials with lower embodied carbon, in this case softwood timber, in the wall component was the principle factor in the difference found. Atypical for the UK, timber was not only used as the main structural material but also as an external cladding material, rather than the more traditional brick cladding. The displacement of brick cladding with larch cladding produced a carbon saving of 24%.

The use of a timber frame also produced a lighter weight structure when compared with a masonry cavity construction. The lighter structure required less substructural support and, consequently, reduced foundation materials. Again reducing the use of high embodied carbon materials, such as concrete and steel reinforcing.

A further factor contributing to the lower embodied carbon of the case study home is the efficiency of volume production associated with MMC.

Despite efficiencies in the production of the MMC frame, onsite waste production in this case study was still a significant factor in the total embodied carbon, 14% of the total. However the manufacturing of the frame was a relatively small contributor to total waste related CO_2 produced, just 4% of waste related embodied carbon (Fig. 7). This suggests further reductions in embodied carbon can be made by both increasing the amount of manufacturing that occurs off site and by reducing the amount of waste that occurs on site.

On small sites, such as those typical to rural sites, any 'waste' or surplus materials from unused contingency, over ordered materials or sizeable offcuts, are produced in relatively small quantities with poor separation of waste.

However, anecdotal evidence from informal conversations and observation of site operations during the construction of the case study suggests significant barriers, such as time, lack of local infrastructure and health and safety legislation, exist to hinder the reuse of these materials locally or recycled back into the supply chain.

The waste data collected, from both MMC and the on-site construction, was not of sufficient quality to make a robust quantification. Further research is needed to quantify the resource efficiency claims from MMC methods in comparison with that from onsite construction.

Despite the high proportion of timber throughout the structure half of the materials related embodied carbon was found to be associated with the construction of the substructure, foundations and ground floor (Fig. 5). The relative importance of these substructural components reduces with the increase of carbon intensive materials in other components, for example in scenario 3 the proportion attributed to these elements is lower at 35%. This suggests that these sub structural elements and the materials used would be a suitable target for reducing the embodied carbon still further in such MMC timber framed houses.

The substructural components were comprised of cementitious rich materials and bricks. Bricks, unless they are unfired, have a high embodied carbon factor. The cement in concrete and mortars can have a high energy input during manufacture and, consequently a relatively high embodied carbon, in addition to the release of CO_2 during the chemical changes that take place during manufacture. The amount of embodied carbon associated with cement production depends upon the primary materials and the energy source used in its production. The emissions associated with cement production can be reduced by the displacement of fossil fuels with both renewable energy and waste materials as energy sources.

Reducing the environmental burdens from MMC timber frame construction further could be achieved in two ways. Firstly, reducing the use of cement by substituting with lower embodied carbon alternatives. Materials include using ground granulated blast furnace slag, fly ash and other pozzolanic materials or lime based materials.

Secondly, using design strategies to reduce the volumes of cement required. These could include removing the oversite concrete 'raft', using isolated point foundations rather than strip foundations or using steel helical screw piles. Although a relatively high embodied energy product steel helical screw piles are both reusable and recyclable.

Both these strategies would radically reduce the use of carbon intensive materials where no additional benefit to their use is possible in lightweight construction. There would also be other additional benefits such as reduced earthworks requiring less spoil and waste material for export off site, lower energy inputs and further benefits at end of life deconstruction. These strategies could be equally applied to traditional masonry construction.

However, it is too simplistic to consider embodied carbon as an isolated issue. Cement, and the concrete that contains it, is the main material used in the global construction industry. For example, concrete materials have a high thermal mass that can assist in reducing occupational heating and cooling energy loads of up to 23% [25] if used appropriately [26]. However, in the case study, typical of most timber frame construction, the majority of its useful mass is isolated within the structure or beneath other material finishes and, consequently, unavailable for the useful thermal storage that could offset the environmental burdens of its manufacture. The full lifecycle, including occupation, maintenance and end of life deconstruction and disposal needs to be considered. Consideration of embodied carbon needs to be integrated at the earliest design stage. If environmental burdens are to be minimised sensibly whilst maximising additional benefits there needs to be systemic intelligent thought in building design.

8. What are the implications for the UKs national carbon targets?

If it is assumed that the average area of a new home in the UK is 91 m^2 [27] and that the targeted 3 million new homes will be new construction, rather than replacing existing stock, there will be an additional 273 million m^2 of new housing at a rate of 240,000 new homes per year. The carbon consequences will depend significantly upon how these homes are constructed. The range could be between 110 and 167 $MtCO_2$ depending on the proportions of all timber MMC to traditional masonry construction used. If it is assumed that the new homes are constructed in the same way as scenario 1 and have an embodied carbon of 405 $kgCO_2$ per m^2, the carbon consequences of the construction of 3 million new homes in terms of the UK's carbon emissions would be approximately 110 $MtCO_2$. However, this is unlikely to occur. Whilst the market share of timber frame new homes is approximately 22% [28] the predominant preferred cladding material is brick and the majority of housing construction (including multi occupancy buildings such as flats) continues to be masonry. Therefore if only traditional build occurred the carbon consequences would be approximately 167 $MtCO_2$ (assuming an embodied carbon of 612 $kgCO_2$ per m^2 as estimated in scenario 3).

To put this into context the UK currently emits 542.6 $MtCO_2$, of which 142.2 $MtCO_2$ (or 30%) is attributable to residential energy use [8]. On an annual basis the embodied carbon of construction of 240,000 homes could be between 6 and 10% of the annual housing emissions. A total of 3 million new homes could equal or exceed the annual emissions of the total housing stock.

The drive towards zero carbon by 2016 will negate a proportion of this increase through reduced energy demand in use, if they are replacing existing stock rather than adding to it. Whether this reduced demand will offset the increased embodied carbon required for construction will depend upon the materials used, the technologies used to supply services, the demands of the inhabitants and other social pressures and the end of life deconstruction.

In addressing carbon mitigation the UK's policy focus on energy efficiency and clean energy, to the exclusion of embodied carbon, may be missing an important point in terms of global carbon emissions. Much of the embodied carbon occurs elsewhere, materials are often produced and imported from elsewhere and these emissions are unaccounted for. However, carbon emissions and the environmental damage of these emissions are no respecter of administrative borders.

9. Conclusions

The results indicate that the embodied carbon of a house constructed using offsite panellised timber frame (scenario 1) is approximately 35 tCO_2. A comparative model of an equivalent home constructed using traditional masonry construction (scenario 3) was found to have an embodied carbon of 52 tCO_2, 51% greater.

Despite timber being the predominant structural and cladding material, concrete is the most significant material by proportion in embodied carbon terms, responsible for 36% of materials related embodied carbon. Much of this is embodied in the substructure.

In considering the construction as a whole further embodied carbon savings can be made by:

- increased offsite manufacturing of components
- consideration of material specification and selection of sustainable materials or materials with reduced environmental impact (e.g. cement substitutes)
- design and placement of materials within the structure (such as mass materials accessible as thermal storage)
- On-site waste minimisation strategies

A systemic lifetime approach is also needed. Decision making based on a single issue, such as embodied carbon, can be misleading and counterproductive in the long run. The example given is of concrete; as a material it does have a large embodied energy and carbon burden, however it is a useful material and can also act to reduce the occupational energy demand if it is employed strategically within a structure.

And finally to answer the question posed, there will indeed be very significant carbon consequences to the UK's house building programme, despite its aspiration to 'zero' carbon status. The authors estimate between 110 and 167 $MtCO_2$ depending on the proportions of all timber MMC to traditional masonry. A significant proportion of this will be outside the national accounting framework and, consequently, concealed within imported materials and products. The overall impact will be dependent upon the types of construction employed and how integrated the sustainable construction agenda is embedded along the whole supply chain, from inception through design and on to construction, occupation and deconstruction.

Acknowledgments

The study on which this paper is based was funded by Carbon Connections and Broadland District Council. The authors thanks are given to Ms Jo Franklin of John Youngs Homes and David Powis of Space 4 for their time and help in collecting the necessary data for this study.

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Definitions

Embodied carbon: the CO_2 emissions produced during the extraction of resources, transportation, manufacture, assembly, disassembly and end of life disposal of a product. In construction the majority of CO_2 is produced from the burning of fossil fuels. Significant amounts of CO_2 are also released through chemical conversion processes during the manufacture of cement. Embodied carbon is given as kg or tonnes of CO_2.

Embodied energy: the total primary energy required for the extraction of resources, transportation, manufacture, assembly, disassembly and end of life disposal of a product. Embodied energy is given as megajoules (MJ) or gigajoules (GJ) of energy.

Glulam beams: (or glue laminated timber) are structural timber beams comprised of several layers of dimensioned timber glued together. Both these forms of engineered wood products represent an efficient use of available timber by making use of smaller, less desirable and waste timber.

I-beam: are structural beams with an I or H shaped cross section. The horizontal flanges are formed from solid timber (usually softwood) to which a vertical web (typically a plywood) is jointed and glued. Lighter, using cheaper low grade materials and as structurally strong as expensive solid timber.

Modern methods of construction (MMC): are synonymous with off-site manufacturing and prefabrication of building components and modules in factory settings, including complete buildings. MMC is used to describe a range of technologies and processes involving prefabrication and off-site assembly of building components and readymade rooms (modules).

Primary energy: refers to the energy contained in raw fuels that have not been subjected to any conversion or transformation process. It refers to the sum of energy (including that used in extracting raw materials, manufacture, transport) of delivered energy. Primary energy is given as megajoules (MJ) or Gigajoules (GJ).